

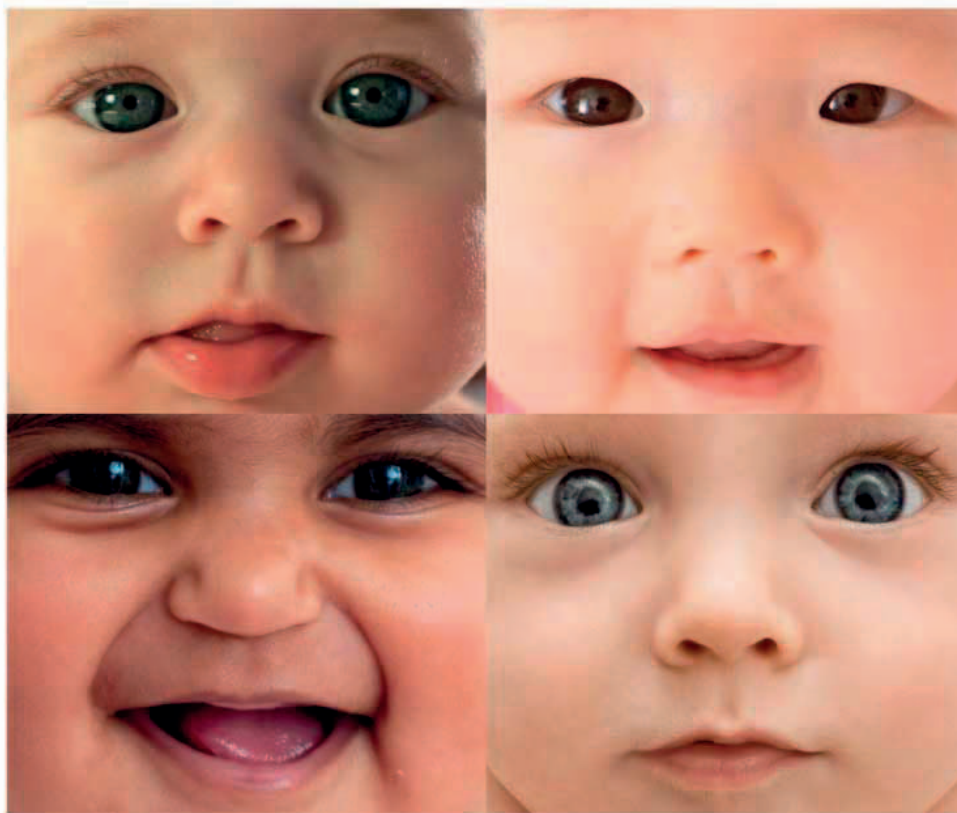
Q Which member of my family is my newborn baby going to look like most?

Your baby might have a range of features and characteristics from different members of each of your families—an aunt's hair, a cousin's sense of humor, or your mother's nose. How DNA and inheritance work can help to explain this.

Your baby shares half her DNA with you and half with your partner. In turn, you share half of your genes with each of your parents. This means that your baby shares a quarter of her genes with your parents—her grandparents.

Through each generation genes are shuffled and reshuffled. At conception, the embryo receives 23 chromosomes from the mother's egg and 23 chromosomes from the father's sperm. These pair up to make a total of 46 chromosomes. The chromosomes contain the genes that you inherit. When the sperm fertilizes the egg, chromosomes join together and randomly exchange genes.

Likewise, even though full siblings inherit genes from the same parents, these aren't necessarily the same genes, nor the same dominant alleles. Remember, you've each passed one of two copies of every gene you have to your baby, but because the two copies of each of your genes aren't identical (every copy has variations), you won't always pass the same version of each gene to each child. Mix this up for all the genes from both of you, and it turns out that the permutations are endless.



Blue eyes, brown eyes, different shaped eyes, and noses Humans are amazingly diverse. This is due to both genetic inheritance and variation in environmental factors.

Q What determines whether I have a boy or a girl?

Your baby's gender is determined by specific chromosomes—the tubelike structures that contain her unique genetic blueprint. Healthy human beings have 23 chromosome pairs; one chromosome in each pair is inherited from each parent. The 23rd pair of chromosomes is called the sex chromosome and it determines what sex a baby will be. A woman's egg always contains the female chromosome, known as X and a man's sperm contains either an X (female) or Y (male) chromosome. An embryo that has two X chromosomes will be female, whereas one with an XY combination will be male. It is therefore the man's sperm that determines whether a baby is male or female.

Q Can environmental factors influence how my children look if they are the same sex?

Children who grow up in the same household are never in the same environment at every moment. They could have different bedrooms, different teachers, and experience different emotions. Also, how tall each child grows is partly influenced by the food they eat. One child may have started solids with only carrots and pears, but your second is more likely to start with the foods that her big sister's now eating. What you eat plays its part, too. If your taste preferences have changed since the first pregnancy, the nutritional influence on each of your daughters will be different—and that can impact their physical characteristics.

Q Can my baby's genes determine whether or not she will be overweight later on in her life?

No, but they may predispose her to weight gain if it runs in the family. In this case, whether or not your baby turns out to be overweight will depend a lot on environmental factors. Her eating habits, stress levels, and enjoyment of sports, can all influence whether or not the genes for gaining excess weight switch on. How you eat in pregnancy can influence it as well. Your baby can taste what's in the amniotic fluid, so if she gets used to sweeter flavors, say, while she's inside you, it may be more difficult to encourage her to eat vegetables as she gets older.